

■ ■ Global Fashion Retail

Sales Analysis — Power BI Portfolio Project

6.4 M	€733 M	1.6 M	35
Transactions	Total Revenue	Customers	Stores
7 Countries	3 Categories	17,940 Products	Jan 2023 – Mar 2025
US · UK · France Germany · Spain Portugal · China	Masculine · Feminine Children	Across all categories and sub-categories	Full 26-month date range

Tools & Skills

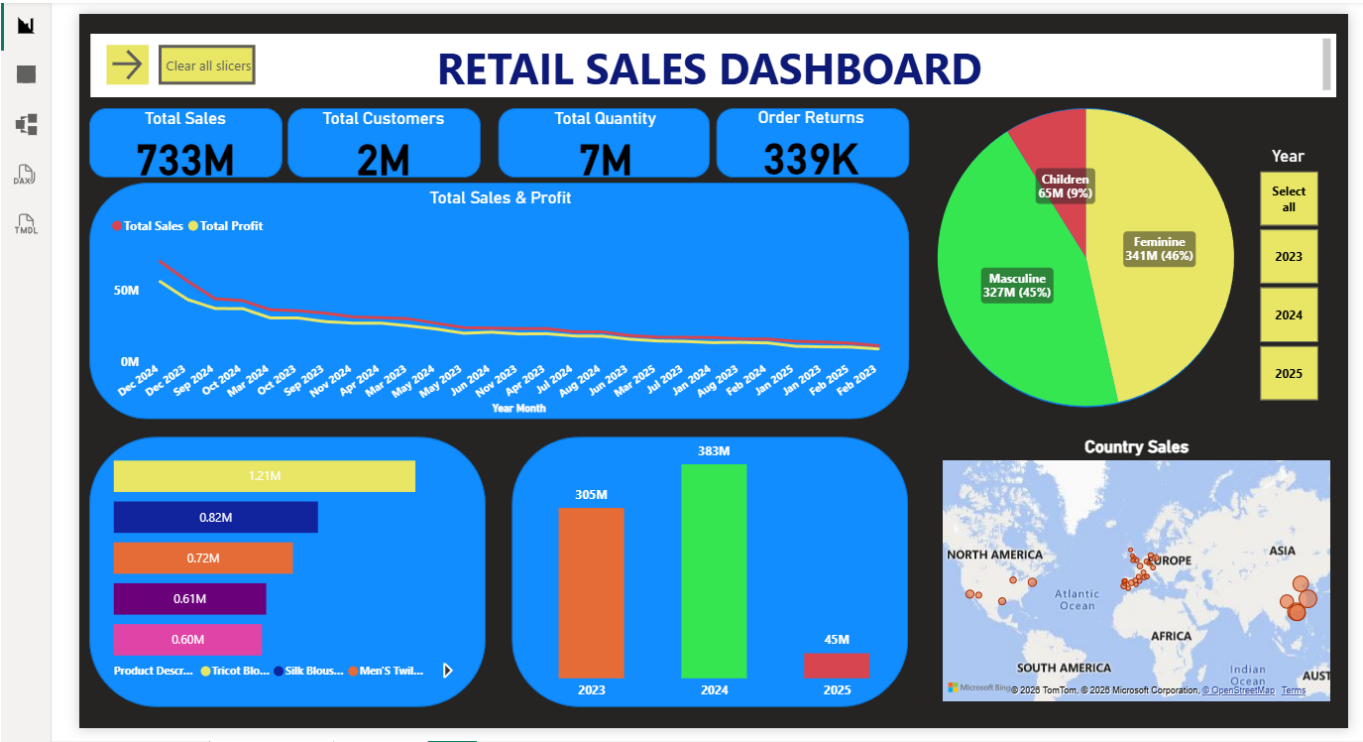
Power BI Desktop	Power Query (M)	DAX	Star Schema	Git / GitHub
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End-to-end project: Raw CSV import → Data cleaning → Data modelling → DAX measures → Interactive report

Dataset: Global Fashion Retail Stores — Kaggle | Author: Sumit

Page 1 — Retail Sales Dashboard

Executive overview: KPI cards, monthly trend, YoY column chart, category breakdown, top products, and a world map with precise store coordinates.



Dashboard page — overview for executives and stakeholders

Key Insights from this page

- Revenue grew 25%+ YoY: €305M (2023) → €383M (2024)
- Feminine category leads at 46% of total sales, Masculine close behind at 45%
- 2025 shows only Jan–Mar data (€45M) — full year projection exceeds 2024
- Store map shows geographic spread across US, Europe, and China

Page 2 — Deep Dive Report View

Granular analysis for business analysts: monthly MoM pivot, country-level table with 10 metrics per row, and dual slicers (Year + Country).

REPORT VIEW

Total Sales

733M

AVG ORD VALUE

161

TOTAL PROFIT

610M

PROFIT MARGIN

83%

YoY GROWTH

7%

Select all

2023

2024

2025

Country

All

Sales By Year & Month

Year	2023			2024			2025		
Year Month	Total Sales	Sales PM	MoM Growth %	Total Sales	Sales PM	MoM Growth %	Total Sales	Sales PM	MoM Growth %
Apr 2023	23,411,150.47	30,250,631.08	-23%						
Apr 2024				31,001,218.04	36,558,622.07	-15%			
Apr 2025							17,418,769.72		-100%
Aug 2023	16,166,451.30	17,220,268.30	-6%						
Aug 2024				21,010,104.95	21,040,931.40	-0%			
Dec 2023	56,652,073.45	23,427,145.70	142%						
Dec 2024				70,613,680.45	31,526,169.60	124%			
Total	305,002,250.38	248,350,176.93	23%	382,961,152.81	368,999,545.81	4%	44,885,004.10	115,498,684.55	-61%

Year	Country	Total Sales	% Total Sales	Sales LY	YoY Growth	Total Profit	Profit Margin	Total Quantity	Total Returns	Return Rate	Total Customer	Repeat Cu
2023	Deutschland	13,092,006.14	2%		0.00%	6,838,120.45	52.23%	357747	17091	5.26%	1643306	
2023	España	9,664,612.51	1%		0.00%	5,092,547.94	52.69%	262880	12833	5.38%	1643306	
2023	France	11,621,145.33	2%		0.00%	6,072,388.22	52.25%	318285	15448	5.34%	1643306	
2023	Portugal	11,598,728.90	2%		0.00%	6,097,171.29	52.57%	315862	15302	5.33%	1643306	
2023	United Kingdom	8,089,780.91	1%		0.00%	2,829,688.15	34.98%	300962	14403	5.27%	1643306	
2023	United States	33,436,109.55	5%		0.00%	18,695,874.71	55.92%	841237	40413	5.28%	1643306	
2023	中国	217,499,867.04	30%		0.00%	205,374,976.92	94.43%	693739	32885	5.22%	1643306	
2024	Deutschland	14,821,204.10	2%	13,092,006.14	13.21%	7,778,279.76	52.48%	409763	19784	5.31%	1643306	
2024	España	11,816,579.65	2%	9,664,612.51	22.27%	6,281,556.44	53.16%	323585	15632	5.32%	1643306	
2024	France	12,361,384.29	2%	11,621,145.33	6.37%	6,549,072.95	52.98%	340074	16233	5.25%	1643306	
Total		732,848,407.29	100%	687,963,403.19	6.52%	610,490,688.47	83.30%	7059187	338829	5.28%	1643306	

Dashboard

Report

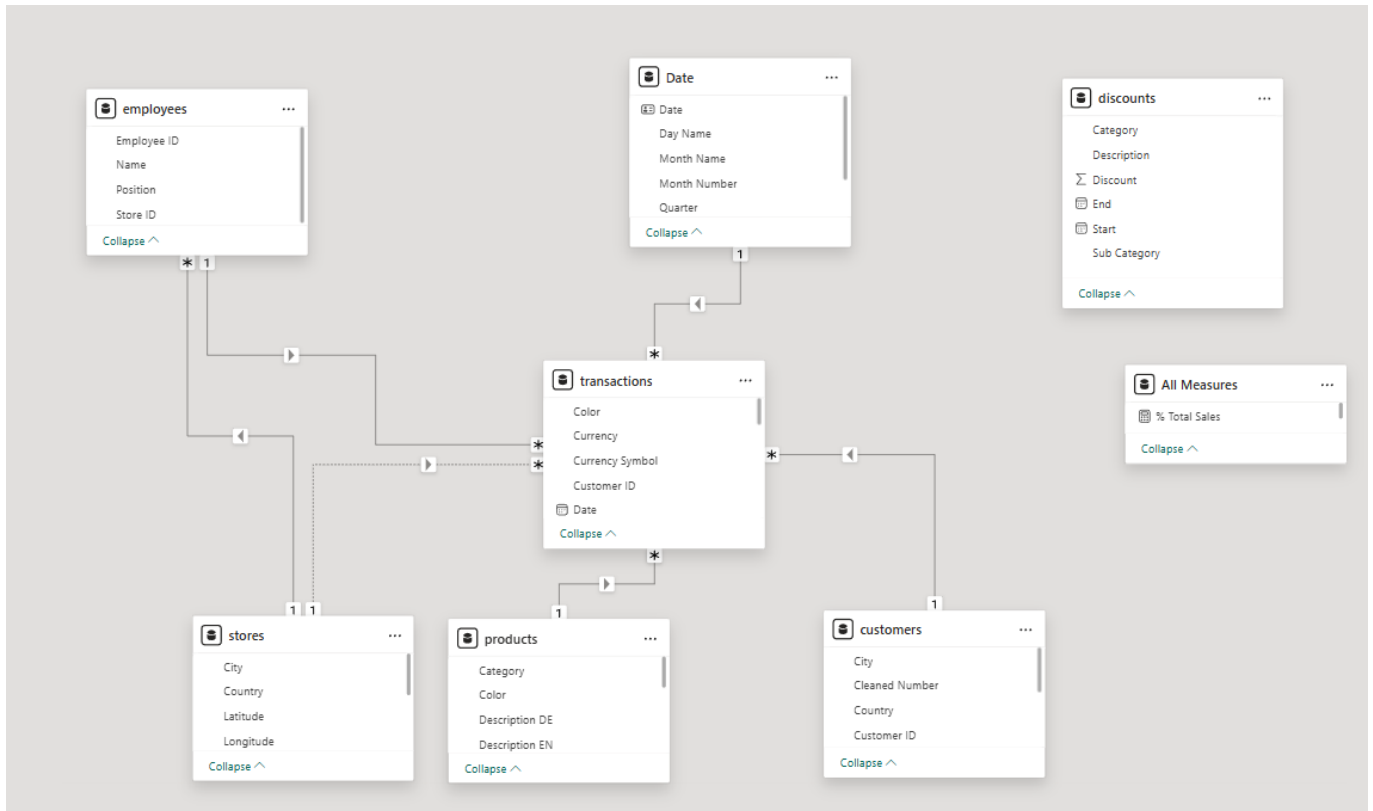
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Report page — detailed country and monthly breakdown

Country	2024 Total Sales	YoY Growth	Profit Margin	Return Rate
Deutschland	€14.8M	+13.2%	52.5%	5.3%
España	€11.8M	+22.3%	53.2%	5.3%
France	€12.4M	+6.4%	53.0%	5.3%
China	€217.5M	—	94.4%	5.2%
United States	€33.4M	—	55.9%	5.3%

Data Model — Star Schema

Industry-standard star schema with Transactions as the Fact table at the center, surrounded by 5 dimension tables. Discounts connects via DAX (no direct key).



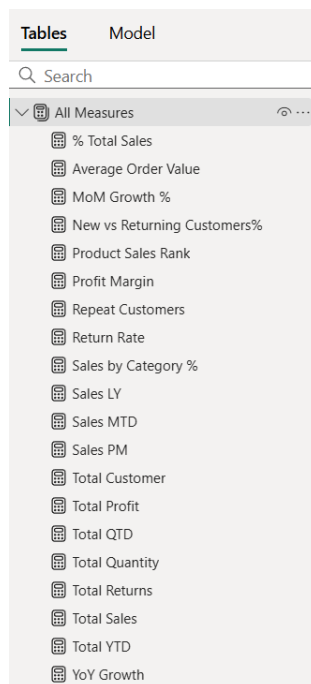
Star schema — Transactions fact table with 5 dimension tables

Relationships

From (Many side)	To (One side)	Key Column	Note
transactions	customers	Customer ID	
transactions	products	Product ID	
transactions	stores	Store ID	
transactions	employees	Employee ID	
transactions	Date table	Date Only	Calculated col — strips time from Date/Time
employees	stores	Store ID	Snowflake branch
discounts	(none)	—	Connected via DAX date-range logic only

DAX Measures Reference

All measures stored in a dedicated 'All Measures' table. 20 measures across 5 categories.



All Measures table in Power BI Fields pane

1. Core Measures

Measure	DAX Formula	Purpose
Total Sales	SUM(transactions[Line Total])	Base revenue — includes returns (negative)
Total Profit	SUM(transactions[Profit])	Sum of per-row profit calculated column
Total Quantity	SUM(transactions[Quantity])	Units sold
Total Customer	DISTINCTCOUNT(transactions[Customer ID])	Unique buyers
Total Returns	CALCULATE(COUNTROWS(transactions), transactions[Transaction Type]="Return")	Return count via CALCULATE filter

2. Ratio & Rate Measures (always use DIVIDE — never /)

Measure	DAX Formula	Key concept
Profit Margin	DIVIDE([Total Profit], [Total Sales], 0)	DIVIDE handles zero-division
Return Rate	DIVIDE([Total Returns], COUNTROWS(transactions), 0)	% of all rows that are returns
Average Order Value	DIVIDE([Total Sales], DISTINCTCOUNT(transactions[Invoice ID]), 0)	\$ per invoice, not per item
% Total Sales	DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(transactions)), 0)	ALL() removes filters for denominator
Sales by Category %	DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(products[Category])), 0)	% within category only

3. Time Intelligence (all reference 'Date'[Date], not transactions[Date])

Measure	DAX Formula	Note
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Sales LY	<code>CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))</code>	Last year same period
Sales PM	<code>CALCULATE([Total Sales], DATEADD('Date'[Date], -1, MONTH))</code>	Previous month
Sales MTD	<code>TOTALMTD([Total Sales], 'Date'[Date])</code>	Month-to-date running total
Total QTD	<code>TOTALQTD([Total Sales], 'Date'[Date])</code>	Quarter-to-date
Total YTD	<code>TOTALYTD([Total Sales], 'Date'[Date])</code>	Year-to-date
YoY Growth	<code>DIVIDE([Total Sales] - [Sales LY], [Sales LY], 0)</code>	% growth vs same period last year
MoM Growth %	<code>DIVIDE([Total Sales] - [Sales PM], [Sales PM], 0)</code>	% growth vs previous month

4. Ranking Measures

Measure	DAX Formula	Note
Product Sales Rank	<code>RANKX(ALL(products[Description EN]), [Total Sales])</code>	ALL() compares against every product, ignoring current filter context

5. Customer Behaviour Measures

Measure	DAX Formula	Pattern used
Repeat Customers	<code>CALCULATE(DISTINCTCOUNT(transactions[Customer ID]), FILTER(VALUES(transactions[Customer ID]), CALCULATE(DISTINCTCOUNT(transactions[Invoice ID])) > 1))</code>	FILTER over VALUES — iterate per customer, test condition, then count
New vs Returning Customers%	<code>DIVIDE([Repeat Customers], [Total Customer], 0)</code>	Combines two measures

Key DAX Rules Learned in This Project

- Use `DIVIDE()`, never `/` operator — avoids divide-by-zero crashes in visuals
- Time intelligence functions must point to the Date TABLE column, not the fact table's date
- Turn OFF Auto Date/Time — it creates hidden hierarchies that silently corrupt time intelligence
- `CALCULATE` is the most important DAX function — it overrides filter context
- Calculated Column vs Measure: fixed per-row property → Column | changes with filters → Measure
- `Transactions[Date]` was Date/Time — created Date Only = `DATEVALUE(Transactions[Date])` for the relationship key